nileWays

Comprehensive Graduation Project Documentation — Part I

Digital Egypt Pioneers Initiative
React Front-End Web Developer Track

Project Name	nileWays - Smart Tourism & Travel Platform	**Team Leader**	Mahmoud Hatem Mahmoud
Project Type	Team Graduation Project	**Team Size**	5 Members

GitHub Repository: https://github.com/Mahmoud72207/nileways-project.git

1. PROJECT PROPOSAL & GOVERNANCE

1.1 Executive Summary & Scope

nileWays is an integrated web platform engineered to organize and digitize a tourist's journey across Egypt. From airport arrival transportation to discovering historic sites, navigating local transport via a Cairo Metro planner, comparing hotel rates, and booking curated tours, nileWays unifies the entire Egyptian travel experience into a single responsive web interface.

Problem Statement
Tourists face unverified local taxi pricing, scattered hotel booking engines lacking local transparency, confusing transit systems, and static agency tour packages.

Proposed Solution
A unified web application with 13 specialized pages featuring an interactive Metro route planner, real-time hotel price tiers, custom itineraries, and direct tour bookings.

1.2 Project Milestones & Schedule

Phase	Period	Milestone Deliverables
Phase 1	4 Oct – 17 Oct 2026	Proposal, architecture design, Figma UI wireframes, repo setup.
Phase 2	18 Oct – 24 Oct 2026	Database schema, RESTful API contracts, shared React components.
Phase 3	25 Oct – 7 Nov 2026	Frontend view development (All 13 routes & state management).
Phase 4	8 Nov – 21 Nov 2026	API integration, feature freeze, cross-browser testing & validation.
Phase 5	22 Nov – 4 Dec 2026	Deployment, responsiveness audit, final presentation & report.

1.3 Team Member Roles & Responsibilities

Member Name	Assigned Role	Core Technical Responsibilities
Mahmoud Hatem Mahmoud	Leader / Front-End	App Architecture, Routing, Auth flow, Home & Favorites views.
Ahmed Waleed Mohamed	Front-End Developer	Metro Planner, Transport UI, Interactive Map & Tours views.
Nouran Ashraf Yassin	Front-End Developer	Hotels directory, Hotel Details, Places & Place Details pages.
Tasbeeh Mahmoud Mohamed	UI/UX & Front-End	Design System, Wireframes, My Bookings, Shared UI (Nav/Footer/404).
Kirollos Ghobrial Faheem	Back-End Developer	Database schemas, RESTful APIs, Admin API endpoints, Testing.

2. SYSTEM APPLICATION ARCHITECTURE & ROUTE MATRIX

The nileWays client interface consists of **13 distinct pages** designed to guide the tourist through every step of their trip:

#	Page Route	Owner	Page Function & User Experience Scope
1	`/` (Home)	Mahmoud	Hero search, landing categories, featured places, quick transport selector, call-to-actions.
2	`/places`	Nouran	Filterable catalog of tourist destinations (Historical, Natural, Culture, Coastal).
3	`/places/:id`	Nouran	Comprehensive location info, opening hours, ticket pricing, photo gallery, nearby stays.
4	`/tours`	Mahmoud	Curated package tours, filters by duration, price, and theme (Day trips, Nile cruises).
5	`/tours/:id`	Mahmoud	Detailed tour itinerary, included amenities, guide info, and direct booking form.
6	`/metro-planner`	Nouran	Interactive Cairo Metro route calculation, line interchange alerts, and ticket cost calculator.
7	`/hotels`	Kirollos	Hotel directory with rating, distance to landmarks, price ranges, and amenity filters.
8	`/hotels/:id`	Kirollos	Hotel details, room availability preview, guest reviews, and booking initiation.
9	`/map`	Ahmed	Full-screen interactive map showing nearby places, metro stations, and saved stops.
10	`/auth` (Login/Register)	Ahmed	User authentication modal/page with validation, JWT session handling, and social login UI.
11	`/favorites`	Kirollos	Saved places, bookmarked hotels, and custom travel lists for quick offline retrieval.
12	`/my-bookings`	Tasbeeh	User dashboard managing active, completed, and canceled transport/tour/hotel reservations.
13	* (404 Not Found)	Tasbeeh	Custom error fallback page with friendly redirect navigation back to primary pages.

3. WEEKLY DEVELOPER SCHEDULE (W1 – W8)

Dev	Weeks 1-2 (Foundation)	Weeks 3-4 (Core UI)	Weeks 5-6 (Advanced Pages)	Weeks 7-8 (Integration)
Mahmoud	Setup Vite, React Router, Tailwind	Auth UI, Login/Register JWT handling	Home Page UI & Favorites logic	Global State, Deployment, Routing QA
Ahmed	UI Components & Icons setup	Metro Planner & Station Data logic	Tours Catalog & Tour Booking view	Interactive Leaflet Map integration
Nouran	API Client & Mock data JSON	Places Catalog & Category Filters	Hotels Search & Price Comparison UI	Place/Hotel Detail pages API hookup
Tasbeeh	Figma Design System & Tokens	Navbar, Footer & Theme Switcher	My Bookings Dashboard UI	404 Page, Responsiveness & UX Polish
Kirollos	Express server & Mongo/Postgres DB	Auth endpoints & Places APIs	Hotels, Tours & Metro route endpoints	Bookings API, Security & Postman Docs

4. USER STORIES & DETAILED USE CASES

US-01: Metro Transit Route Calculation
As a tourist exploring Cairo,
I want to enter my departure and destination stations,
So that I can see the exact metro line transfers, duration, and ticket price.

US-02: Transparent Hotel Price Tiering
As a budget-conscious traveler,
I want to compare hostels and luxury hotels on one page,
So that I can pick accommodation matching my exact budget.

US-03: Tour Package Direct Booking
As a short-stay visitor,
I want to browse guided day trips and book immediately,
So that I do not waste time dealing with unverified street agencies.

US-04: Saved Favorites & Bookings
As a registered user,
I want to save places to my Favorites and track my active bookings,
So that I can access my trip itinerary on the go.

5. FUNCTIONAL REQUIREMENTS (FR-01 TO FR-10)

ID	Requirement Title	Target Route / Module	Detailed Functional Specification
FR-01	User Auth & Profile	`/auth`	User login/registration with form validation and JWT token persistence.
FR-02	Global Search	`/`, `Navbar`	Instant keyword search across Places, Hotels, and Tour itineraries.
FR-03	Place Filtering	`/places`	Multi-attribute filter by city (Cairo, Luxor, Aswan), category, and tags.
FR-04	Metro Route Calculation	`/metro-planner`	Algorithm calculating shortest path, line switches (Line 1/2/3), and fare.
FR-05	Hotel Rate Comparison	`/hotels`	Sorting by price per night, star rating, proximity to sights, and user reviews.
FR-06	Tour Reservation	`/tours/:id`	Date picker, group size selection, price calculation, and reservation post.
FR-07	Interactive Mapping	`/map`	Map view rendering markers for places, hotels, and metro stations.
FR-08	Bookmarking Engine	`/favorites`	Toggle favorite status on cards with instant update in LocalStorage/Backend.
FR-09	Booking Management	`/my-bookings`	Display user reservations with cancellation and download voucher support.
FR-10	Fallback Route Handler	* (404 Page)	Catch invalid route attempts and provide explicit return paths.

6. NON-FUNCTIONAL REQUIREMENTS & GIT WORKFLOW

Non-Functional Requirements

- **Performance:** Sub-2s LCP load time; optimized image lazy-loading.
- **Responsiveness:** Fluid layouts for 320px mobile up to 4K displays.
- **Accessibility:** WCAG 2.1 AA compliance with high-contrast text.

Version Control & Branch Strategy

- `main`: Production-ready code only.
- `dev`: Staging branch for integrated feature testing.
- `feature/*`: Individual branches for each member's pages.